increased by 40% in the next year and by 50% in the following year. Calculate the average rate of increase in overhead expenses over the three years.

Solution:

In averaging ratios and percentages, geometric mean is more appropriate. Let us consider X represents Expenses at the end of the year.

% Rise	X	$\log X$
32	132	2.1206
40	140	2.1461
50	150	2.1761
		$\Sigma \log X = 6.4428$

Table : 8.6

$$GM = Anti\log\left(\frac{\Sigma \log X}{n}\right)$$

$$= Anti\log\left(\frac{6.4428}{3}\right)$$

$$= Anti\log(2.1476)$$

$$GM = 140.5$$

Average rate of increase in overhead expenses

$$140.5 - 100 = 40.5\ \%$$



Geometric mean cannot be calculated if one of the observations is zero.

8.1.5 Harmonic mean

Harmonic mean is defined as the reciprocal of the arithmetic mean of the reciprocal of the individual observations. It is denoted by HM.

Thus, HM $= \dfrac{n}{\left(\dfrac{1}{X_1}+\dfrac{1}{X_2}+.....+\dfrac{1}{X_n}\right)}$

When the number of items is large the computation of harmonic mean in the above manner becomes tedious. To simplify calculations we obtain reciprocals of the various items from the tables and apply the following formulae:

(i) In individual observations

$$HM = \frac{n}{\left(\dfrac{1}{X_1}+\dfrac{1}{X_2}+.....+\dfrac{1}{X_n}\right)} \quad \text{(or)}$$

$$HM = \frac{n}{\Sigma\left(\dfrac{1}{X}\right)}$$

where n is number of observations or items or values.

(ii) In discrete frequency distribution

$$HM = \frac{N}{\Sigma\left(\dfrac{f}{X}\right)} \text{ where}$$

N = total frequency= Σf

(iii) In continuous frequency distribution

$$HM = \frac{N}{\Sigma\left(\dfrac{f}{m}\right)}$$

Where m is midpoint and N is total frequency

Example 8.9

Calculate the Harmonic Mean of the following values:

1, 0.5, 10, 45.0, 175.0, 0.01, 4.0, 11.2.

08_11th_BM-STAT_Ch-8-EM.indd 185 21-04-2020 12:26:05 PM